

10.01 Interpreting graphs

Name _____

Class _____

Artwork by Marni Tuala · We acknowledge the Bundjalung people, the Traditional Custodians of the land on which we learn.

Learning intentions

We are learning to:

- read and interpret picture graphs, column graphs, sector graphs and line graphs
- explain what a graph tells us about real data

Success criteria

I can:

- ☐ use a **key** and a **scale** to read values from a graph
- ☐ estimate amounts from a **sector graph** using fractions of a circle
- ☐ compare values and give reasons for patterns in a graph

WE DO

Grey boxes: work **with your teacher**. Copy the working from the board.

YOU DO

Blue boxes: work **on your own**. Show your working in the box.

Key

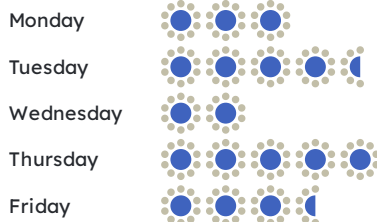
Scale

Sector graph

Notes

Level 1 · Getting started

Books borrowed from the library



● = 4 books

WE DO

With your teacher: copy what they write on the board.

Example 1

How many books were borrowed on Tuesday?

Example 2

How many more books on Thursday than Wednesday?

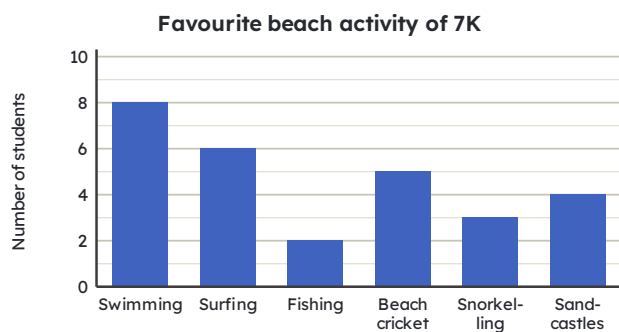
YOU DO

On your own: Set A. Use the library picture graph above.

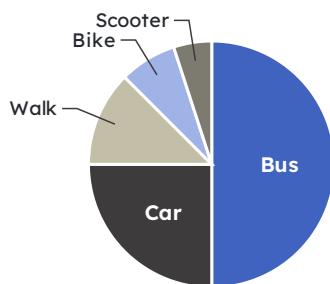
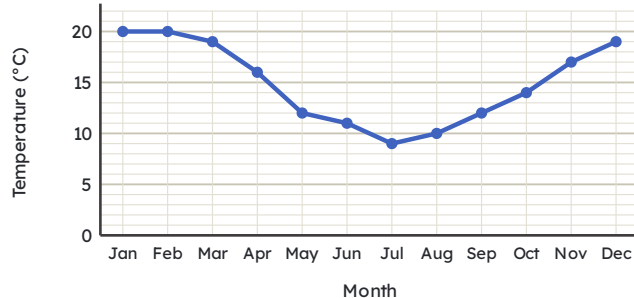
- | | |
|--|---|
| 1 What does 1 symbol stand for? <input type="text"/> | 6 Which day had the most books? <input type="text"/> |
| 2 What does $\frac{1}{2}$ a symbol stand for? <input type="text"/> | 7 Which day had the fewest books? <input type="text"/> |
| 3 How many books on Monday? <input type="text"/> | 8 How many more on Friday than Monday? <input type="text"/> |
| 4 How many books on Wednesday? <input type="text"/> | 9 How many on Monday and Tuesday? <input type="text"/> |
| 5 How many books on Friday? <input type="text"/> | 10 How many symbols show 16 books? <input type="text"/> |

YOU DO

On your own: Set B. Use the column graph.



- 1 How many chose surfing?
- 2 How many chose fishing?
- 3 Which was the most popular?
- 4 Which was the least popular?
- 5 Which activity had 5 votes?
- 6 How many chose snorkelling?
- 7 How many more swam than fished?
- 8 How many chose surfing or fishing?
- 9 How many students are in 7K?
- 10 What does 1 gridline stand for?

**Level 2 · Building up****How 80 Year 7 students travel to school****Mean minimum temperature, Kingscliff****WE DO**

With your teacher: copy what they write on the board.

Example 3

What fraction of the students catch the bus? How many students is that?

Example 4

Which month is coldest? How much colder is June than March?

YOU DO

On your own: Set C. Questions 1–5 use the sector graph and 6–10 use the line graph.

- 1 Which way of travel is the most common?

- 2 Estimate how many students travel by car.

- 3 Estimate how many students walk.

- 4 True or false: fewer than 10 ride a bike.

- 5 How many students do **not** catch the bus?

- 6 Which 2 months are warmest? What is their mean minimum?

- 7 Name 2 months with the same mean minimum.

- 8 Which month has a mean minimum of 14 °C?

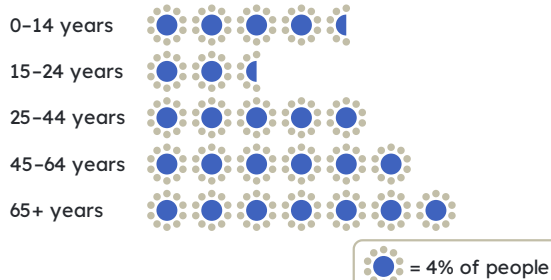
- 9 Which 2 months in a row have the biggest rise?

- 10 Which 2 months in a row have the biggest drop?

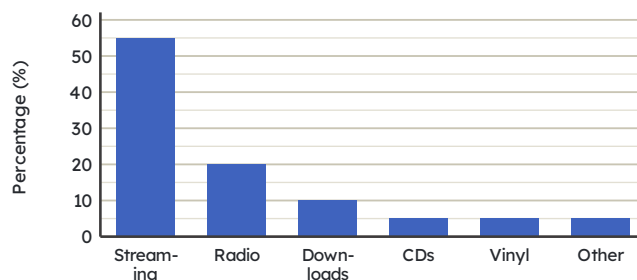
Level 3 · Challenge

Finished Set D? Try the extension on page 4.

Ages of people in a coastal town



How Australians listen to music

**WE DO** With your teacher: copy what they write on the board.**Example 5** What percentage of the town is 45+? How many is that out of 5000?**Example 6** What % stream music? How might this look different 20 years ago?**YOU DO** On your own: Set D. Questions 1–5 use the picture graph and 6–10 use the column graph.

1 What percentage of people are aged 65+?

2 What percentage of people are under 25?

3 The town has 5000 people. How many are aged 0-14?

4 How many symbols would show 30% of the town?

5 Why might 65+ be the biggest group in this town?

6 Which 2 ways of listening add up to 75% of people?

7 What fraction of people listen to the radio?

8 In a survey of 400 people, how many use downloads?

9 Out of 400 people, how many more stream than use radio?

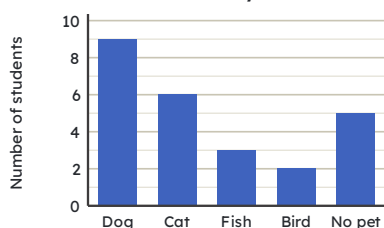
10 How might this graph change in 10 years? Why?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Pets owned by 7B

**Level 1**

How many students in 7B own a cat?

Level 2

What fraction of the class owns a dog?

Level 3

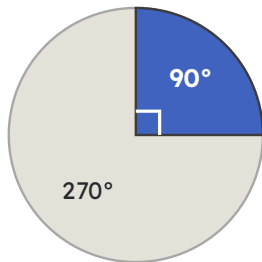
What percentage of 7B has no pet? Out of 200 Year 7s, how many would you expect to have no pet?

Extension · Stretch yourself

Worked example

In a survey of 72 students, the **Soccer** sector of a sector graph has an angle of 90° . How many students chose soccer?

1. A full circle is 360° , so 90° is $90/360 = \frac{1}{4}$ of the circle
2. $\frac{1}{4}$ of 72 = $72 \div 4 = 18$



90° out of 360°
 = $\frac{1}{4}$ of the circle
 = $\frac{1}{4}$ of the students

Answer: 18 students chose soccer

YOU DO

On your own: read the worked example, then try these.

- E1** A survey of 240 people has a **Tennis** sector of 60° . How many chose tennis?

- E2** In a survey of 180 people, 45 chose pizza. What angle should the pizza sector be?

- E3** In a picture graph, one symbol = 8 students. How many students do $3\frac{3}{4}$ symbols show? How would you show 20 students?

Lesson summary

Missed the lesson? Start here.

Reading any graph

1. Read the **title**: what is the graph about?
2. Check the **key** or the **scale** on each axis.
3. Use a ruler to line up bars and points with the scale.
4. Answer with the right **units** (students, $^\circ\text{C}$, %).

Level 1

In a picture graph, = 10 students. A row shows $3\frac{1}{2}$ symbols. How many students is that?

1. 3 full symbols = $3 \times 10 = 30$
2. Half a symbol = $10 \div 2 = 5$
3. $30 + 5 = 35$

Answer: 35 students

Level 2

A sector graph shows 80 people. The **Surfing** sector is a quarter of the circle. How many chose surfing?

1. A quarter means divide by 4
2. $80 \div 4 = 20$

Answer: 20 people

Level 3

A graph shows 45% of 400 students catch the bus. How many students is that?

1. 45% means 45 out of every 100
2. 400 has 4 hundreds, so $45 \times 4 = 180$

Answer: 180 students

Extension

A 120° sector shows the results for 90 people. How many people is that?

1. $120/360 = \frac{1}{3}$ of the circle
2. $\frac{1}{3}$ of 90 = $90 \div 3 = 30$

Answer: 30 people

This space is left blank on purpose. It is the back of your exit ticket.